

Cover letter

[AUTHOR INPUT REQUIRED: submission date]

Dear Editor,

Please consider our Original Research article, “Return-aware reinforcement learning with a Pointer policy for resource-constrained mountain-road inspection planning,” for publication in *Engineering Applications of Artificial Intelligence*.

The manuscript addresses an engineering problem in which an unmanned aerial vehicle must select priority-labelled fixed inspection points along mountain roads while retaining sufficient energy, distance, time and dynamic feasibility for safe return. Its artificial-intelligence contribution is a sequence-aware actor–critic that combines proximal policy optimization, a Pointer policy, priority-biased candidate encoding and a return-aware multi-resource feasibility mask. Its engineering contribution is a frozen, reproducible comparison across unseen procedural maps, Copernicus digital-surface-model simulations, conventional planners and two classes of perturbation.

The study reports 21,648 audited route evaluations. The proposed model did not universally maximize coverage: ant-colony, simulated-annealing and mixed-integer planners achieved higher coverage in some settings, but with longer online planning times. The principal evidence instead shows near-identical final coverage to an advantage-actor–critic–Pointer comparator, substantially higher coverage than fixed-slot multilayer-perceptron proximal policy optimization, more favourable post-hoc training stability and sample efficiency, and a large loss when return reserve was removed. We believe this balanced result is directly aligned with EAAI’s interest in validated artificial-intelligence applications, real-time engineering use, safety and reliability.

The submission is single-column and prepared for double-anonymized review. The anonymized manuscript contains no author-identifying information; the title page is separate. The abstract explicitly distinguishes the artificial-intelligence contribution from the engineering application. A reproducibility package contains the frozen protocol, results, statistical source data, figure source data, environment information and hashes. Copernicus assets are handled under their current attribution and access conditions.

This work has not been published previously and is not under consideration elsewhere. All authors have approved the manuscript and agree with its submission. [AUTHOR INPUT REQUIRED: verify these declarations.] Competing interests and funding are reported on the separate title page. The authors will review and approve the generative-AI disclosure before submission.

Thank you for your consideration.

Sincerely,

[AUTHOR INPUT REQUIRED: corresponding author name, affiliation and contact details]